Quadratics Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Quadratic formula: the −b

ORIGINAL PROBLEM

Solve $x^2 - 6x + 5 = 0$ with the quadratic formula.

INCORRECT WORKED SOLUTION

$a = 1, b = -6, c = 5$

$$x = \frac{-6 \pm \sqrt{36 - 20}}{2}$$

$$x = \frac{-6 \pm 4}{2}$$

$x = -1$ or $x = -5$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Substituting c

ORIGINAL PROBLEM

How many real solutions does $2x^2 + 3x - 2 = 0$ have? Solve it.

INCORRECT WORKED SOLUTION

$a = 2, b = 3, c = 2$

$b^2 - 4ac = 9 - 4(2)(2) = -7$

The discriminant is negative: no real solutions.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Quadratics Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

3. Square root method

ORIGINAL PROBLEM

Solve $(x - 2)^2 = 25$

INCORRECT WORKED SOLUTION

$(x - 2)^2 = 25$

$x - 2 = 5$

$x = 7$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Completing the square

ORIGINAL PROBLEM

Solve $x^2 + 6x = 7$ by completing the square.

INCORRECT WORKED SOLUTION

$x^2 + 6x + 36 = 7 + 36$

$(x + 3)^2 = 43$

$x = -3 \pm \sqrt{43}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Quadratics Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

5. Finding the vertex

ORIGINAL PROBLEM

Find the vertex of $y = x^2 - 4x + 1$.

INCORRECT WORKED SOLUTION

$a = 1, b = -4$

$x = \frac{b}{2a} = \frac{-4}{2} = -2$

$y = (-2)^2 - 4(-2) + 1 = 13$

Vertex: $(-2, 13)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Reading vertex form

ORIGINAL PROBLEM

Identify the vertex of $y = (x + 3)^2 - 2$.

INCORRECT WORKED SOLUTION

$y = (x + 3)^2 - 2$

$h = 3, k = -2$

Vertex: $(3, -2)$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Quadratics Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

7. Opening direction

ORIGINAL PROBLEM

Find the vertex and tell whether it is a maximum or minimum for $y = -2x^2 + 8x$.

INCORRECT WORKED SOLUTION

$a = -2$

The parabola opens up.

The vertex is a minimum.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Interpreting solutions

ORIGINAL PROBLEM

A ball's height is $h = -16t^2 + 64t$. When does it hit the ground again?

INCORRECT WORKED SOLUTION

Set $h = 0$: $-16t^2 + 64t = 0$

$-16t(t - 4) = 0$

$t = 0$

The ball lands at 0 seconds.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Quadratics Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Simplifying the formula

ORIGINAL PROBLEM

Simplify $x = \frac{-2 \pm \sqrt{12}}{2}$

INCORRECT WORKED SOLUTION

$x = \frac{-2 \pm \sqrt{12}}{2}$

$x = -1 \pm \sqrt{12}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Isolating x^2

ORIGINAL PROBLEM

Solve $2x^2 = 8$

INCORRECT WORKED SOLUTION

$2x^2 = 8$

$x^2 = 16$

$x = \pm 4$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points